Grow-a-gatchi: A Gardening Companion and Helper

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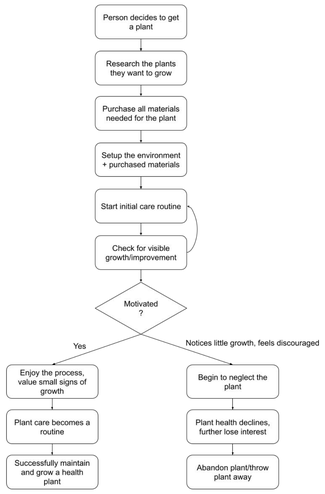
Motivation

People who care for indoor plants often struggle to stay motivated because the rewards of plant growth unfold slowly over time. Existing plant care apps focus on reminders and instructions, but they don't address the loss of motivation that comes from delayed gratification and the lack of immediate feedback, leading to unwillingness to stay consistent.

Although a person decides to start gardening, the path from starting an initial plant care routine to the end of successfully growing a healthy plant is challenged as the fork of motivation, which will either lead to a thriving indoor garden or abandonment.



User Research

We conducted 20+ interviews as a consistent measure of gathering information from our target audience(aspiring indoor gardeners). After the first round of interviews, we chose to conduct a second round of interviews, honing in on the general problem of lack of motivation/consistency expressed by interviewees.

General Consensus(Interviews): Although many did acknowledge price and lack of knowledge as obstacles to maintaining an indoor garden, the majority of people said that the thought of caring daily for plants is a bigger factor in not starting nor maintaining an indoor garden.

Personas

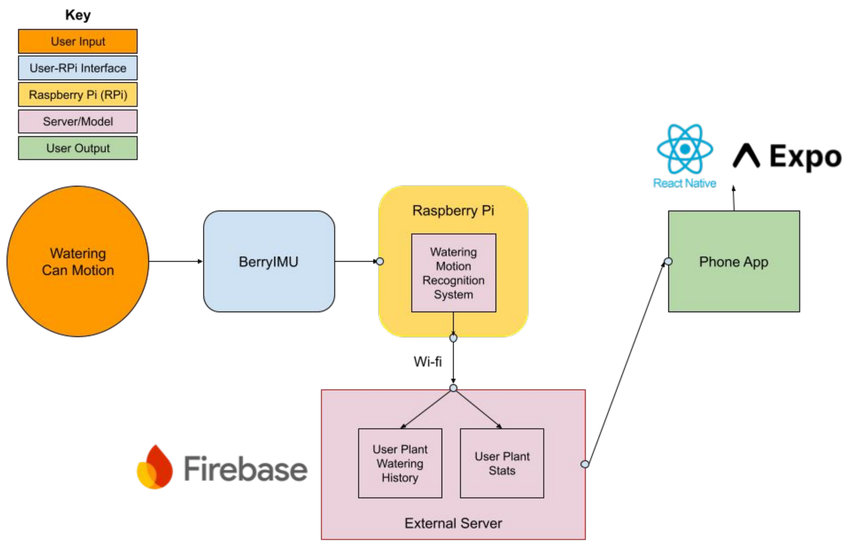
Peter, a busy college junior
He is busy with classes and clubs but wants to explore new hobbies, like indoor gardening

Madison, a successful gardener
She already maintains her own small army of plants

System Overview

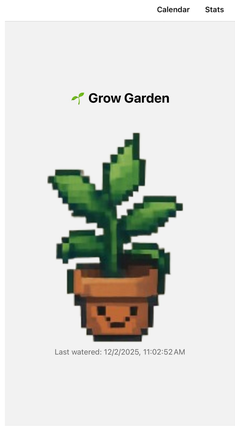
Introducing Grow-a-gatchi! Grow-a-gatchi aims to counter the lack of immediate gratification through linking real-life maintenance of a plant to a virtual space. The users can see their progress through maintaining a plant sprite alongside their real-life counterpart. In addition to the gratification gained by seeing the virtual plant's expression changes and leveling up, the app helps maintain consistency through educating, tracking, and notifying the user of a proper watering schedule. The app aims to provide a non-intrusive solution to assist aspiring gardeners, no matter their current situation.

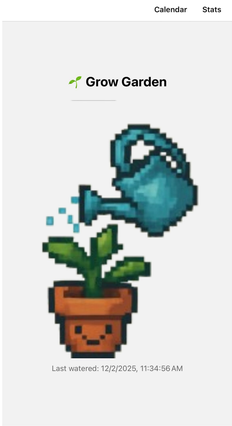
The System Diagram



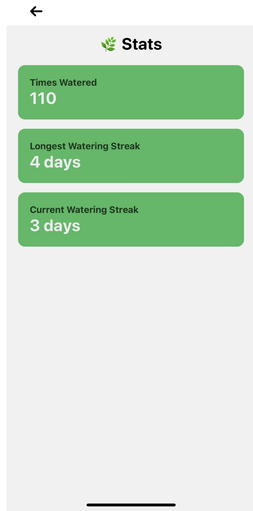
The RPi and IMU are attached to the bottom of a watering can, where they collect acceleration data to determine if the user has indeed watered their plant. If so, the RPi sends a ping to our database. The database processes this ping to create records of watering history as well as stats such as how many times the user has watered their plant.

The Phone App: Main Page with your Grow-a-gatchi





The Phone App: Plant Stats & Calendar



Evaluations

Evaluator Comments
- Users enjoyed a cute tamagotchi that provides immediate feedback
- Calendar + Stats allows users to keep track of when to water their plant and keeps them motivated to maintain their streak!
- Enjoyed how reliable the hardware detected them watering their plant!

Future Directions:
- Community aspects! Grow a communal garden with your friends by placing your virtual plants all together. This helps keep each other accountable + allows you to get inspired from your friends' plants!
- Updated UIs: More plants, a prettier background (your apartment, an outside garden…)
- Education and Stats: Facts and tips to help you take care of your plant beyond watering it